COMMON BAYESIAN OPTIMIZATION POLICIES

7.6 $\sim$ 7.10

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Mutual information and entropy searh

Information-Theoretic Optimization Policies

Entropy Search & Mutual Information

- Use **mutual information (MI)** as the acquisition function.
- Often referred to as **Entropy Search** methods.

General Framework

- Let ω denote an unknown quantity of interest (eg, x^*, f^*).
- Each observation provides information about ω.
- Select observations to maximize **information gain**.

Decision-Theoretic Result

- Using information gain as a utility function leads to:

$$\text{Acquisition} = I(y; \omega)$$

- Different choices of ω produce different policies.

Mutual Information

Let ω and ψ be random variables with $p(\omega)$ and $p(\psi)$. The mutual information between them is

$$I(\omega;\psi) = \iint p(\omega,\psi)\log\frac{p(\omega,\psi)}{p(\omega)p(\psi)}\,d\omega d\psi$$

$$= D_{\mathrm{KL}}[p(\omega,\psi)\parallel p(\omega)p(\psi)]$$

Given observed data $\mathcal{D}$,

$$I(\omega;\psi\mid\mathcal{D}) = \iint p(\omega,\psi\mid\mathcal{D})\log\frac{p(\omega,\psi\mid\mathcal{D})}{p(\omega\mid\mathcal{D})\,p(\psi\mid\mathcal{D})}\,d\omega d\psi$$

- Measures dependence between variables
- Zero when independent, symmetric, nonnegative

Entropy Interpretation of Mutual Information

$$
\begin{aligned}
I(\omega;\psi) &= \iint p(\omega,\psi)\log\frac{p(\omega,\psi)}{p(\omega)p(\psi)}\,d\omega d\psi \\
&= \iint p(\omega,\psi)\log p(\omega|\psi)\,d\omega d\psi - \int p(\omega)\log p(\omega)\,d\omega \\
&= \int p(\psi)\left[\int p(\omega|\psi)\log p(\omega|\psi)\,d\omega\right]d\psi + H[\omega] \\
&= H[\omega] - \mathbb{E}_{\psi}\big[H[\omega|\psi]\big] \\
&= H[\psi] - \mathbb{E}_{\omega}\big[H[\psi \mid \omega]\big]
\end{aligned}
$$

Interpretation

- Expected reduction in uncertainty of ω
- Observing ψ decreases entropy of ω

Information Gain in Bayesian Experimental Design

Mutual information arises naturally in Bayesian optimization policy as the one-step expected information gain.

Two types of information gain

- 1. Entropy reduction

$$u(D) = H[\omega] - H[\omega \mid D]$$

- 2. KL divergence form

$$u(D) = D_{\mathrm{KL}}[p(\omega \mid D) \,\|\, p(\omega)]$$

One-step lookahead yields mutual information as an acquisition function.

$$\alpha_1(x; \mathcal{D}) = \mathbb{E}\big[\, u(\mathcal{D}') \mid x, \mathcal{D} \,\big] - u(\mathcal{D})$$

One-Step Expected Information Gain

Marginal information gain:

$$H[\omega \mid D] - H[\omega \mid D']$$

Expected marginal gain:

$$\begin{aligned}\alpha_{\mathrm{MI}}(x; D) &= H[\omega \mid D] - \mathbb{E}_y[H[\omega \mid D'] \mid x, D] \\ &= I(y; \omega \mid x, D) \\ &= H[y \mid x, D] - \mathbb{E}_\omega[H[y \mid \omega, x, D] \mid x, D]\end{aligned}$$

- Measures how informative a new observation y will be
- Select x that maximizes expected knowledge gain

Mutual Information with x^*

Acquisition

$$\alpha_{x^*}(x; D) = I(y; x^* \mid x, D).$$

Behavior

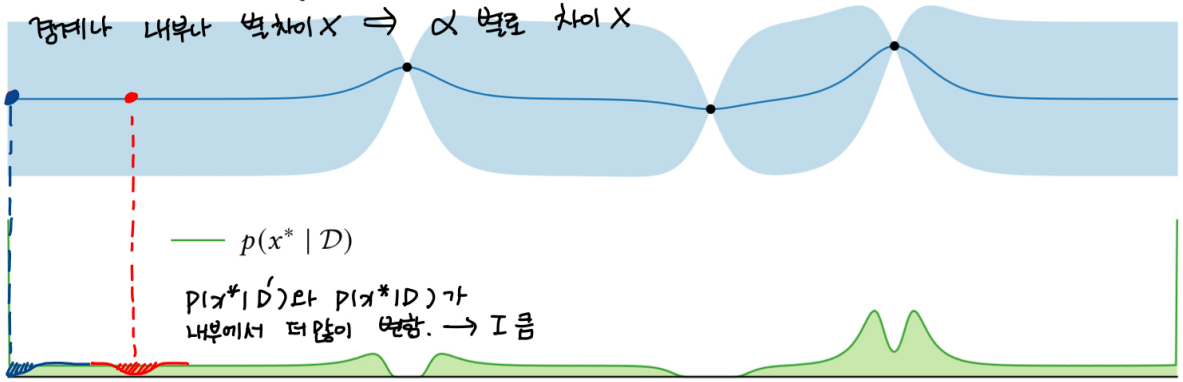
- Tends to explore regions that reduce uncertainty over a **large volume**.
- Even if boundary maximizers are plausible, measuring on the boundary often reveals **less information about** x^*.

Contrast: EI/PI

- EI and PI depend only on the **marginal predictive** $p(y \mid x, D)$.
- Hence they miss **global impact** on beliefs about the optimum.

Visualization

하지만 EI / PI는 $P(y \mid D, x)$ 에만 의존 하는데
경계나 내부나 별 차이 X ⇒ α 별로 차이 X



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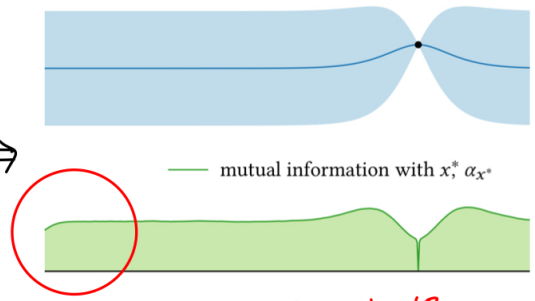


경계에서 acqui~ 값 작음.

Figure 7.13: The posterior belief about the location of the global optimum, $p(x^* \mid \mathcal{D})$, and about the optimum, $p(f^* \mid \mathcal{D})$, for our running example. Note the significant probability mass optimum lying on the boundary.

$$I(y, x^* \mid x, D) = H(x^* \mid D) - E_y(H(x^* \mid D') \mid x, D)$$
$$= E_y(KL(P(x^* \mid D') \parallel P(x^* \mid D)))$$

Mutual Information with f^*

Acquisition

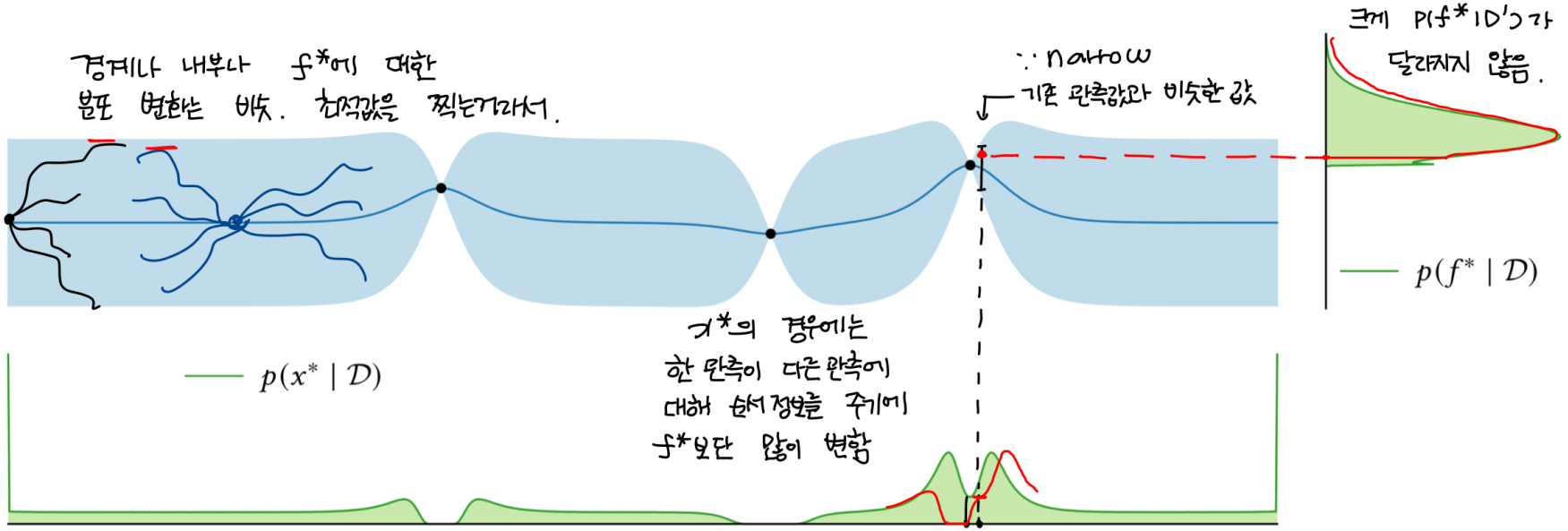
$$\alpha_{f^*}(x; D) = I(y; f^* \mid x, D).$$

- Under exact observations, f^* has **no mass below** the best-seen value (best-seen is a lower bound).

Behavior

- Strongly **penalizes sampling near existing observations**.
- Reason: predictive distributions are **narrow** $\Rightarrow$ little additional information about f^*.
- Boundary points: unlike learning x^*, they can be **similarly informative** about f^* as nearby interior points.

Visualization



Mutual information : x^* vs f^*

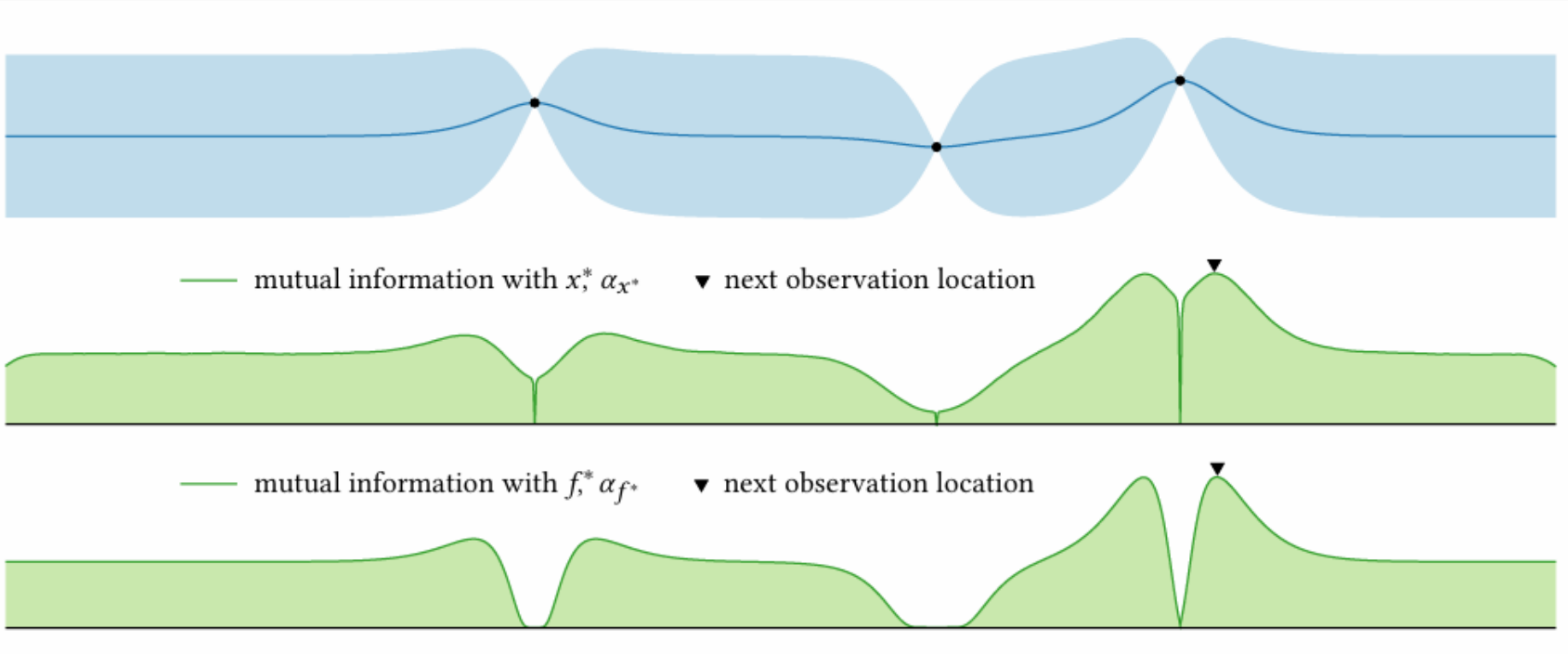


Figure 1: $I(y; x^* \mid x, D)$ and $I(y; f^* \mid x, D)$

- x^*: Boundary obs. reduce uncertainty about f only locally $\Rightarrow$ small change in the dist. of x^* $\Rightarrow$ low mutual information.
- f^*: Sampling near observed points adds little information about the maximum value because variance there is already low.

Q. Why Assume a Unique Global Maximizer?

If the maximizer is not unique:

- x^* becomes a set-valued random variable
- 탐색 공간이 탐색 공간의 모든 가능한 부분집합들을 모아놓은 공간으로 확장
- 표준 Lebesgue measure 기반 density 정의 불가능
- differential entropy 공식을 그대로 적용할 수 없음
- mutual information 을 계산 어려움

Key point: If the maximizer is not unique, entropy and mutual information are not directly defined in the standard framework, making practical computation intractable.

Multi-armed bandits and optimization

Multi-Armed Bandits and Optimization

Multi-armed bandit policies : a model system for sequential decision making under uncertainty.

Setup

- Finite set of arms: $x \in X$
- Reward $y \sim p(y \mid x)$ are stochastic and unknown
- Goal : select a sequence $\{x_i\}$ to maximize cumulative reward Σy_i.
- Expected reward : $\phi(x) = \mathbb{E}[y \mid x]$

Optimal Arm and the Exploration–Exploitation Tradeoff

Optimal arm and value

$$x^* \in \arg\max_{x \in X} \mathbb{E}[y \mid x], \qquad f^* = \max_{x \in X} \mathbb{E}[y \mid x].$$

- If reward distributions were known:
  - Always choose the best arm (trivial optimal policy)
- In reality, rewards are unknown and must be learned.

Exploration vs. Exploitation

- **Exploitation:** choose arm with high expected reward
- **Exploration:** sample uncertain arms to learn

Multi-armed bandit problem is a sequential decision-making under uncertainty that requires balancing exploration and exploitation.

Q. Convergence results in Multi-Armed Bandits

Instantaneous and Cumulative regret

$$r_t = f(x^*) - f(x_t), \quad R_T = \sum_{t=1}^{T} r_t$$

- x^* : optimal arm
- x_t : arm selected at time t

Convergence criterion

$$\frac{R_T}{T} \to 0$$

- 전체 시도 횟수에 대한 평균 후회가 0 으로 수렴한다는 뜻
- 알고리즘이 최적의 선택을 찾아냄을 보장

A. Known Regret Bounds for Bandit Algorithms

Algorithm	Regret Bound
Greedy	$O(T)$
ϵ-greedy (ϵ fixed)	$O(T)$
UCB (Auer et al., 2002)	$O(\log T)$
Thompson Sampling	$O(\log T)$
GP-UCB (Srinivas et al., 2010)	$O^*(\sqrt{T\gamma_T})$
GP-Thompson Sampling	$O^*(\sqrt{T\gamma_T})$

- γ_T : maximum information gain of the Gaussian process
- Sublinear regret $\Rightarrow$ convergence
- 자세한 내용은 10 장을 참고바람

The Bayesian Optimal Policy

Bayesian approach

- Prior over expected rewards: $p(f)$
- Observation model: $p(y \mid x, \phi)$
- Update posterior$D$: $p(f \mid D)$

Optimal τ-step decision rule

$$U(D_\tau) = \sum y_i,$$

$$\alpha_\tau(x; D) = \alpha_1(x; D) + \mathbb{E}\left[\alpha^*_{\tau-1}(D_1) \mid x, D\right],$$

$$x \in \arg\max_{x' \in \mathcal{X}} \alpha_\tau(x'; D).$$

Final decision (pure exploitation)

$$x_\tau \in \arg\max \mathbb{E}[f \mid D_{\tau-1}]$$

Optimization as an Infinite-Armed Bandit

We may model **continuous optimization** as an infinite-armed bandit problem.

$$f : X \to \mathbb{R}, \quad X \text{ infinite}$$

- Each point x is an arm
- Expected reward: $\phi = f(x)$
- Observations: $y \sim p(y \mid x, \phi), \quad \mathbb{E}[y|\phi] = \phi$
- Assumption : Rewards are correlated across the domain

Why Reduce Optimization to a Bandit Problem?

Goal : Optimization : x^* vs Bandit : Σy_i

With many observations, a good policy identifies the best arm.

Regret provides the link

- Rapidly decreasing regret $\Rightarrow$ optimal arm is found
- Infinite-arm case $\Rightarrow$ global optimum is located

Thus, the multi-armed bandit reduction is indeed reasonable.

Maximizing a statistical upper bound

Optimism in the Face of Uncertainty

Optimism in the Face of Uncertainty is a principle that balances exploration and exploitation by assigning a bonus to options with high uncertainty.

Key principle

- Estimate expected reward using available data
- Quantify uncertainty in these estimates

Why?

- Arms with highly uncertain reward have a wide range of plausible values.
- Uncertain arms receive an **exploration bonus**, encouraging exploration of potentially optimal locations.

Upper Confidence Bound (UCB)

Given predictive distribution $p(\phi \mid x, D)$,

$$q(\pi; x, D) = \inf \{\phi' \mid \Pr(\phi \leq \phi' \mid x, D) \geq \pi\}$$

- $q(\pi; x, D)$ is an upper confidence bound on ϕ
- The true value exceeds this bound with probability $1 - \pi$

UCB acquisition function

$$\begin{aligned} \alpha_{\text{UCB}}(x; D, \pi) &= q(\pi; x, D) \\ &= \mu(x) + z_\pi \, \sigma(x) \text{ (if GP)} \end{aligned}$$

- Q. UCB 가 왜 optimism in the face of uncertainty 의 원리가 실제 작동하는지? $\Rightarrow$ 첫번째 항이 exploitation, 두번째 항이 exploration.
- Select where the optimistic estimate is largest.
- Large π $\Rightarrow$ more exploratory

Q. 불확실성에 대한 비관론은 효과적이지 않은가?

불확실성에 대한 비관론 (Pessimism) : 불확실한 선택지를 회피하고, 보수적으로 선택하는 전략

ex) Greedy (continuous)

$$\alpha_{\text{Greedy}}(x; D) = \mu(x)$$

- 탐색 부족, 성능 떨어짐
- 극단적인 경우 exact obs, best seen point 에 갇힘

Comparison of UCB and Mutual info

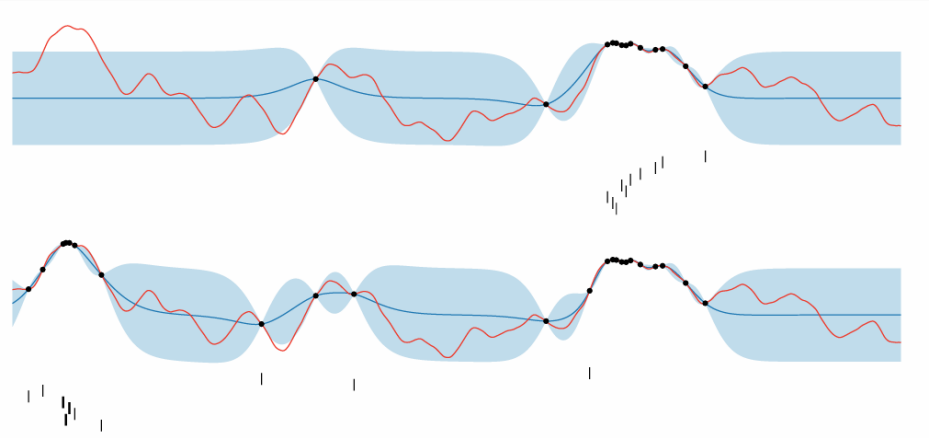


Figure 2: mutual info

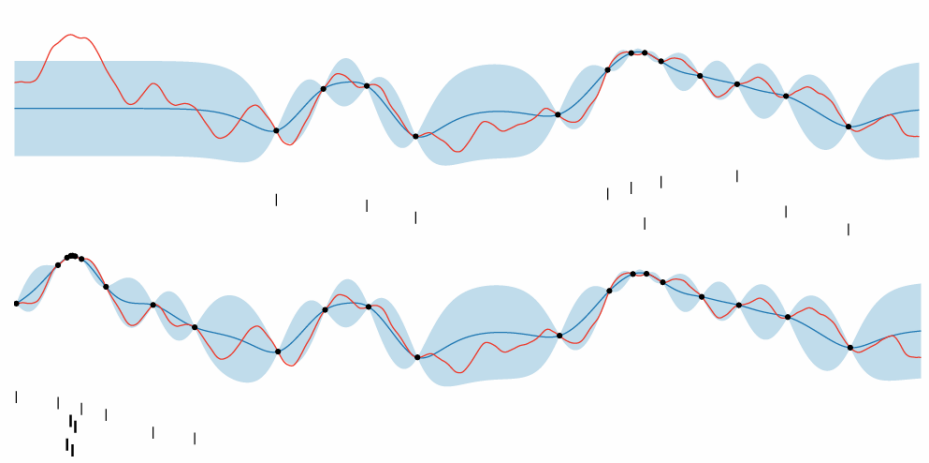


Figure 3: UCB with $\pi = 0.99$

Practical Insights

Parameter tuning

- Typical choices: $\pi \in (0.98, 1)$, or $\pi \approx 0.999$
- Gradually increasing π can maintain exploration as uncertainty decreases.
- [6] prove that if the confidence parameter approaches 1 at an appropriate rate, the policy effectively maximizes the objective at a nontrivial rate. (Q.)

Key relationship (Gaussian process models)

- A point maximizing PI for some threshold τ also maximizes a UCB for some confidence parameter π, and vice versa.
- Therefore, both methods can select identical sets of points.

A . GP-UCB: Theoretical Guarantee (Srinivas et al., 2010)

Policy (GP-UCB).

$$x_t = \arg\max_{x \in D} \; \mu_{t-1}(x) + \sqrt{\beta_t}\, \sigma_{t-1}(x).$$

Regret guarantees (high probability).

- **Theorem 1 (finite D).** If $\beta_t = 2\log\left(\frac{|D| t^2 \pi^2}{6\delta}\right)$, then

$$\Pr\left\{ R_T \le \sqrt{C_1 T \beta_T \gamma_T} \;\; \forall T \ge 1 \right\} \ge 1 - \delta, \qquad C_1 = \frac{8}{\log(1+\sigma^{-2})}.$$

- **Theorem 2 (compact/continuous $D \subset [0, r]^d$).** Under a smoothness condition on GP sample paths and $\beta_t = 2\log\left(\frac{t^2 2\pi^2}{3\delta}\right) + 2d\log\left(t^2 dbr\sqrt{\log\left(\frac{4da}{\delta}\right)}\right)$,

$$\Pr\left\{ R_T \le \sqrt{C_1 T \beta_T \gamma_T} + 2 \;\; \forall T \ge 1 \right\} \ge 1 - \delta, \qquad C_1 = \frac{8}{\log(1+\sigma^{-2})}.$$

Implication. Since $R_T = O\left(\sqrt{T \beta_T \gamma_T}\right)$, the average regret $R_T/T \to 0$ (*no-regret*), so the policy *effectively maximizes* the objective at a nontrivial rate.

PI vs UCB

proof

PI: $\max_x \left\{ 1 - \Phi\left(\frac{\tau - \mu(x)}{\sigma(x)} \right) \right\}, \qquad \phi \mid x, D \sim \mathcal{N}(\mu(x), \sigma^2(x))$

$$\iff \min_x \frac{\tau - \mu(x)}{\sigma(x)}$$

$$\implies \tau = \mu(x^*) + \pi\,\sigma(x^*) \qquad \text{(for optimal } x^*\text{)}$$

$$\implies x^* = \arg\max_x \left[\mu(x) + \pi\,\sigma(x)\right]$$

Otherwise, if $\mu(x') + \pi\,\sigma(x') > \tau$, $\quad \frac{\tau - \mu(x')}{\sigma(x')} < \pi = \frac{\tau - \mu(x^*)}{\sigma(x^*)}$

$\Rightarrow$ contradiction.

Thompson sampling

Thompson Sampling

A stochastic policy for decision making under uncertainty

Next evaluation location is sampled as:

$$x \sim p(x^* \mid D).$$

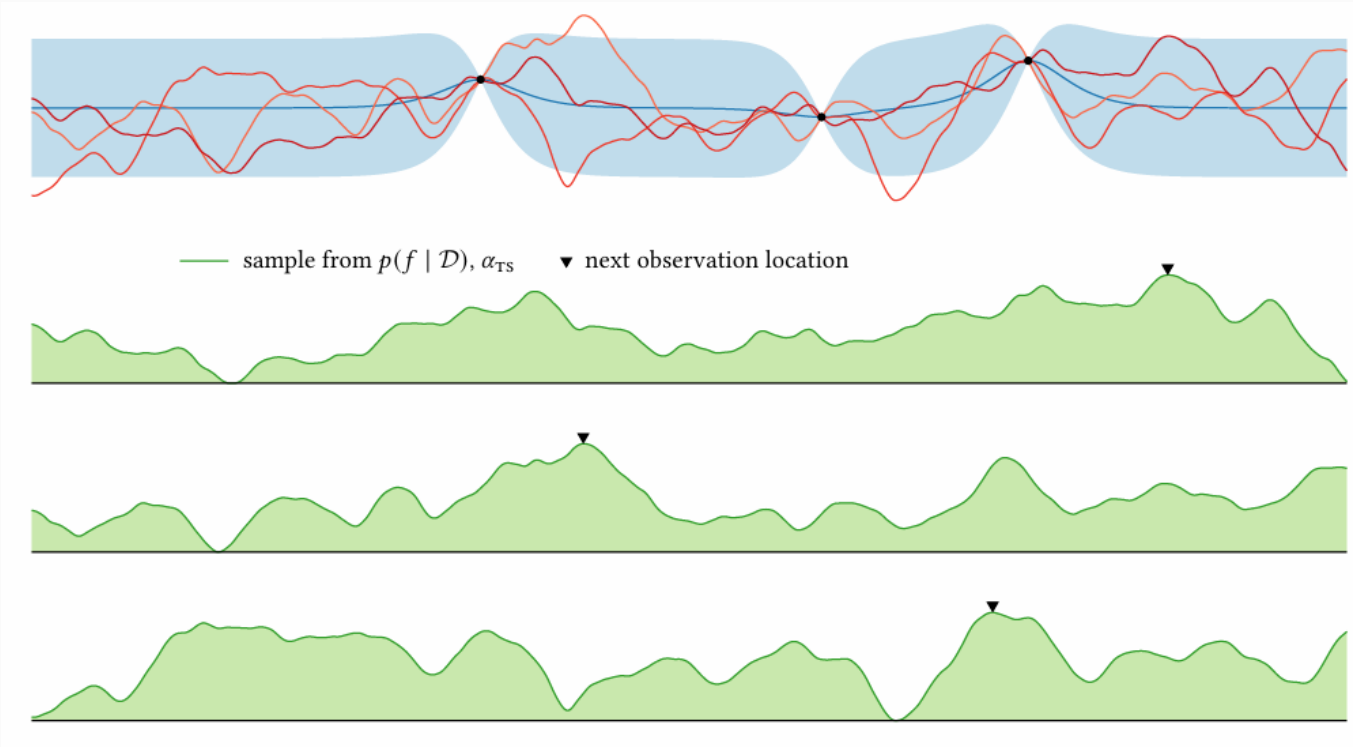
- Maintain a belief about which option is optimal.
- Sample from this belief to guide decisions.
- Sampling proportional to probability of optimality:
  - encourages exploitation
  - guarantees continued exploration
- As a stochastic policy, performance varies across runs.

Random Acquisition Function

Thompson sampling can be interpreted as:

$$\alpha_{\mathrm{TS}}(x; D) \sim p(f \mid D), \quad x \in \arg\max_{x' \in \mathcal{X}} \alpha_{\mathrm{TS}}(x'; D)$$

- Sample a plausible objective function.
- Optimize the sampled function



Q. Can exploration-exploitation be manually controlled in Thompson sampling using parameters?

Possible manual adjustments

- **Posterior variance scaling**

$$f \sim \mathcal{GP}(\mu, \tau^2 K)$$

  $\tau > 1$ increases exploration, $\tau < 1$ favors exploitation.
- **Prior tuning** Changing the prior variance indirectly modifies exploration.
- **ϵ-greedy variants** Occasionally select the greedy action based on the posterior mean.

Practical note

- Requires additional hyperparameter tuning
- Uncertainty decreases with more data $\rightarrow$ less exploration
- Enables safer choices in high-risk applications (e.g., healthcare, autonomous driving)

Q. Finite-Armed Bandit Algorithms

Greedy

$$A_t = \arg\max_a \hat{\mu}_a$$

- Select the arm with the highest empirical mean $\Rightarrow$ no exploration
- Can get stuck at a suboptimal arm

ϵ-Greedy

$$A_t = \begin{cases} \text{random arm} & \text{with prob } \epsilon \\ \arg\max_a \hat{\mu}_a & \text{with prob } 1-\epsilon \end{cases}$$

- Simple way to add exploration

Q. Finite-Armed Bandit Algorithms

UCB (Upper Confidence Bound)

$$A_t = \arg\max_a \left(\hat{\mu}_a + \sqrt{\frac{2 \log t}{N_a}} \right)$$

- Explores arms with high uncertainty (N_a : 해당 arm 선택횟수)

Thompson Sampling

- Posterior sampling (mean reward of arm a) and Arm selection

$$\tilde{\theta}_a \sim p(\theta_a \mid D), \quad A_t = \arg\max_a \tilde{\theta}_a$$

Algorithm	Exploration 방식	특징	Regret
Greedy	없음	단순하지만 실패 위험 큼	$O(T)$
ϵ-greedy	랜덤 탐색	구현 쉬움	$O(T)$ (ϵ fixed)
UCB	uncertainty bonus	deterministic	$O(\log T)$
Thompson Sampling	posterior sampling	stochastic policy	$O(\log T)$

Other ideas in policy construction

Beyond One-Step Lookahead

One-step lookahead is computationally efficient but myopic.

Goal: design less myopic policies that consider future outcomes.

Simple solution: Two-step lookahead

$$\alpha_2(x; D) = \alpha(x; D) + \mathbb{E}\left[\max_{x' \in \mathcal{X}} \alpha(x'; D') \mid x, D\right].$$

ex) Two-step Expected Improvement :

- [5] and [1] demonstrated strong performance.
- Encourages more exploratory behavior early in search.
- But, there is an inherent computational expense
- [3] and [7] developed sophisticated implementation schemes.

Rollout and Nonmyopic Optimization

Rollout approximates optimal policies by simulating future decisions.

- [4] combined EI (base policy) with rollout.
- Also modified the acquisition function by discounting future rewards to balance exploration and exploitation.

$$\alpha_{\tau,\gamma}(x; D) = \alpha_1(x; D) + \gamma\, \mathbb{E}\left[\alpha^*_{\tau-1,\gamma}(D_1) \,\middle|\, x, D\right].$$

Batch rollout

- GLASSES ([2]) : A potential observation is evaluated by the EI gained by simultaneously. evaluating at (that point + batch).
- BINOCULARS ([3]) : Construct a single batch, then select a point from this batch for evaluation.

Artificially encouraging exploration

Another approach is to modify an acquisition function to encourage more exploratory behavior.

Modified Expected Improvement

$$\alpha'_{\mathrm{EI}}(x; D, \epsilon) = \mathbb{E}\left[\max\{\phi - (\phi^* + \epsilon), 0\} \mid x, D\right]$$

- Larger ϵ encourages exploration
- Only available in the case of exact observation.

Other policies

Observation costs can be incorporated via:

$$\alpha'_{\mathrm{EI}}(x; D, \alpha_{sv}) = c(x)$$

Solve for improvement threshold α_{sv} that makes evaluation cost-prohibitive and use α_{sv} as acquisition.

- Low uncertainty region $\Rightarrow$ $\alpha_{sv}(x)$ is small
- behavior similar to location-dependent UCB (Q.)

$$\alpha_{sv} = \mu(x) - g^{-1}(c(x)/\sigma(x))\sigma(x), \quad g(x) = x\Phi(x) + \phi(x)$$

A.

증명)

$$\alpha'_{EI}(x; D, \varepsilon) = E(\max(\phi - T, 0)), \quad T = \phi^* + \varepsilon$$

$$\phi \mid x, D \sim N(\mu(x), \sigma^2(x))$$

$$\alpha'_{EI} = \int_{-\infty}^{\infty} \max(\phi - T, 0) \frac{1}{\sigma(x)} \varphi\left(\frac{\phi - \mu}{\sigma}\right) d\phi, \quad \varphi: \text{표준정규 pdf}$$

$$= \int_{T}^{\infty} (\phi - T) \frac{1}{\sigma} \varphi\left(\frac{\phi - \mu}{\sigma}\right) d\phi, \quad \phi = \mu + \sigma z \text{ 치환}$$

$$= \int_{z_T}^{\infty} (\mu + \sigma z - T) \varphi(z) dz, \quad z_T = \frac{T - \mu}{\sigma}$$

$$= (\mu - T)(1 - \Phi(z_T)) + \sigma \varphi(z_T), \quad \Phi: \text{표준정규 CDF}$$

$$= (\mu - T)\Phi(z) + \sigma \varphi(z), \quad z = -z_T = \frac{\mu(x) - T}{\sigma(x)}$$

$$= \sigma z \Phi(z) + \sigma \varphi(z)$$

$$= \sigma(\varphi(z) + z\Phi(z)), \quad \text{Let } g(z) = z\Phi(z) + \varphi(z)$$

$$= \sigma(x) g(z) = c(x)$$

$$z = g^{-1}(c/\sigma) = g^{-1}(c(x)/\sigma(x)) = \frac{\mu(x) - \phi^* - \alpha_{SV}}{\sigma(x)}$$

$$\alpha_{SV} = \mu(x) - \sigma(x) g^{-1}(c(x)/\sigma(x)) - \underline{\phi^*}$$ x에 상수 → 제거

$$\propto \mu(x) - \sigma(x) g^{-1}\left(\frac{c(x)}{\sigma(x)}\right)$$

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